

1. $A = -2x^2 + 4x + 2$, $B = 2x^2 - 2x + 5$, $C = 3x^2 + 3x + 2$ であるとき, 次の計算をせよ。

(1) $B + C$

答. _____

(2) $3A - 3C$

答. _____

(3) $A - B + C$

答. _____

(4) $A + B + C$

答. _____

(5) $3A - 2B + 2C$

答. _____

(6) $3A - 3B + 3C$

答. _____

(7) $A - 2B + C$

答. _____

(8) $3A + 2B + C$

答. _____

2. 次の 1 次方程式を解け。

(1) $-5x + 4 = -8x - 7$

答. _____

(2) $-7x - 1 = 4x - 9$

答. _____

(3) $2 + 3(x - 7) = -8x$

答. _____

(4) $-9 - 3(-x - 7) = -9x$

答. _____

(5) $6 + 4(-4x - 6) = -x$

答. _____

(6) $5x - \frac{3}{2} = \frac{3}{2}x + 8$

答. _____

(7) $-8x - \frac{1}{2} = 2x + 1$

答. _____

(8) $-8x + \frac{5}{2} = \frac{1}{4}x - \frac{1}{4}$

答. _____

1. $A = -2x^2 + 4x + 2$, $B = 2x^2 - 2x + 5$, $C = 3x^2 + 3x + 2$ であるとき, 次の計算をせよ。

(1) $B + C$

答. $5x^2 + x + 7$

(2) $3A - 3C$

答. $-15x^2 + 3x$

(3) $A - B + C$

答. $-x^2 + 9x - 1$

(4) $A + B + C$

答. $3x^2 + 5x + 9$

(5) $3A - 2B + 2C$

答. $-4x^2 + 22x$

(6) $3A - 3B + 3C$

答. $-3x^2 + 27x - 3$

(7) $A - 2B + C$

答. $-3x^2 + 11x - 6$

(8) $3A + 2B + C$

答. $x^2 + 11x + 18$

2. 次の1次方程式を解け。

(1) $-5x + 4 = -8x - 7$

$3x = -11$

答. $x = -\frac{11}{3}$

(2) $-7x - 1 = 4x - 9$

$-11x = -8$

答. $x = \frac{8}{11}$

(3) $2 + 3(x - 7) = -8x$

$2 + 3x - 21 = -8x$

$11x = 19$

答. $x = \frac{19}{11}$

(4) $-9 - 3(-x - 7) = -9x$

$-9 + 3x + 21 = -9x$

$12x = -12$

答. $x = -1$

(5) $6 + 4(-4x - 6) = -x$

$6 - 16x - 24 = -x$

$-15x = 18$

答. $x = -\frac{6}{5}$

(6) $5x - \frac{3}{2} = \frac{3}{2}x + 8$

両辺に2をかけて, $10x - 3 = 3x + 16$

答. $x = \frac{19}{7}$

(7) $-8x - \frac{1}{2} = 2x + 1$

両辺に2をかけて, $-16x - 1 = 4x + 2$

答. $x = -\frac{3}{20}$

(8) $-8x + \frac{5}{2} = \frac{1}{4}x - \frac{1}{4}$

両辺に4をかけて, $-32x + 10 = x - 1$

答. $x = \frac{1}{3}$